Address AI Bias with Fairness Criteria

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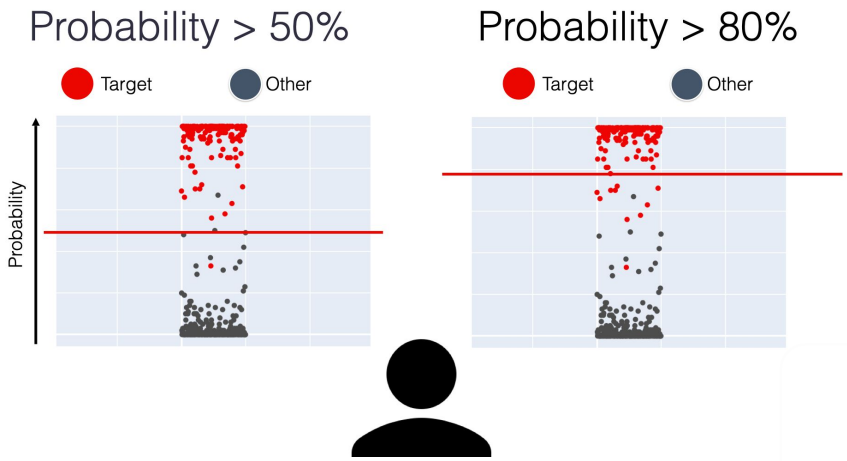
Model performance metrics

Threshold classifiers



A probabilistic classification model returns a number from 0 to 1 for each object.

You then have to decide at which probability you **convert this prediction to a label**.

https://www.evidentlyai.com/classification-metrics/explain-roc-curve

Model performance metrics

correct predictions out of total predictions

$$\text{accuracy} = \frac{TP+TN}{total} = \frac{75+35}{150}$$

Among all the positive predictions of the model how many were true

$$\text{precision} = \frac{TP}{pred.+} = \frac{75}{90}$$

How many were classified by the model correctly among all the true positives

$$\text{recall/sensitivity/TPR} = \frac{TP}{total+} = \frac{75}{100}$$

$$\text{specificity/TNR} = \frac{TN}{total-} = \frac{35}{50}$$

$$F1 = 2 \times \frac{precision \times recall}{precision + recall}$$

$$FPR = \frac{FP}{total-} = 1 - \text{TNR} = \frac{15}{50}$$

$$\text{miss rate/FNR} = \frac{FN}{total+} = \frac{25}{100}$$

$$\text{prevalence} = \frac{total+}{total} = \frac{100}{150}$$

$$\text{misclassification rate} = \frac{FP+FN}{total} = \frac{15+25}{150}$$

	Predicted: -	Predicted: +	
Actual: -	TN=35	FP=15	total - = 50
Actual: +	FN=25	TP=75	total + = 100
	pred. - =60	pred. + = 90	total = 150

TP : true positives
TN : true negatives
FP : false positives
FN : false negatives

TPR: true positive rate
TNR: true negative rate
FPR: false positive rate

$$pred.+ = TP + FP$$
$$total+ = TP + FN$$
$$total- = TN + FP$$

A model that recalls very well but is not very precise may pick up all positive instances correctly, but give too many false positives.

How to measure the balance?

F1 score (0,1): Harmonic mean of Precision and recall. Higher is better.

$$H = n / (1/x_1 + 1/x_2 + \ldots + 1/x_n)$$

Why harmonic mean?
- punishes large differences.

https://jonathan-hui.medium.com/address-ai-bias-with-fairness-criteria-tools-9af1ab8e4289

Threshold classifiers

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

		Predicted: Spam	Predicted: Not
Actual	Spam	800 (TP)	100 (FN)
Actual	Not	500 (FP)	8600 (TN)

Threshold: 0.5

		Predicted: Spam	Predicted: Not
Actual	Spam	600 (TP)	300 (FN)
Actual	Not	100 (FP)	9000 (TN)

Threshold: 0.8

		Predicted: Spam	Predicted: Not
Actual	Spam	200 (TP)	700 (FN)
Actual	Not	10 (FP)	9090 (TN)

Threshold: 0.95

Higher threshold: "more conservative" model, typically lower recall: detect fewer examples of the target class overall.

Lower threshold: "less strict" model, typically higher recall: detect more examples of the target class. May also lead to lower precision.

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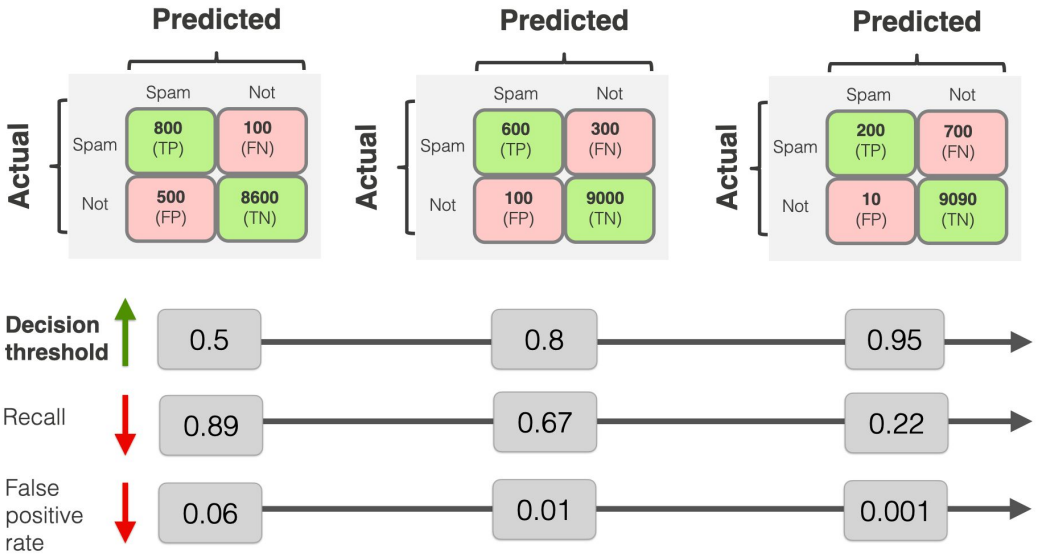
Threshold classifiers

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{FPR} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$



The higher the recall (TPR), the higher the rate of false positive errors (FPR). The lower the recall, the fewer false alarms the model gives.

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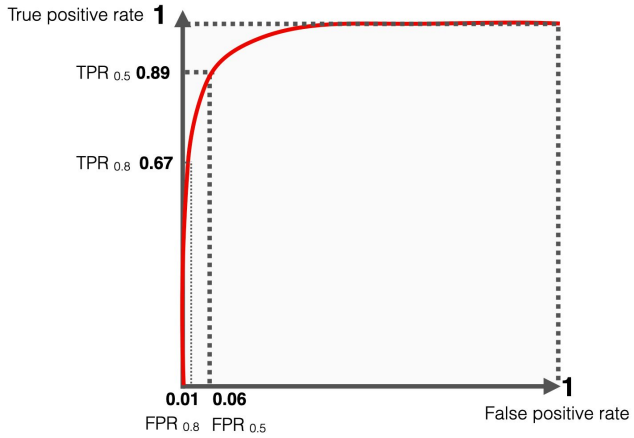
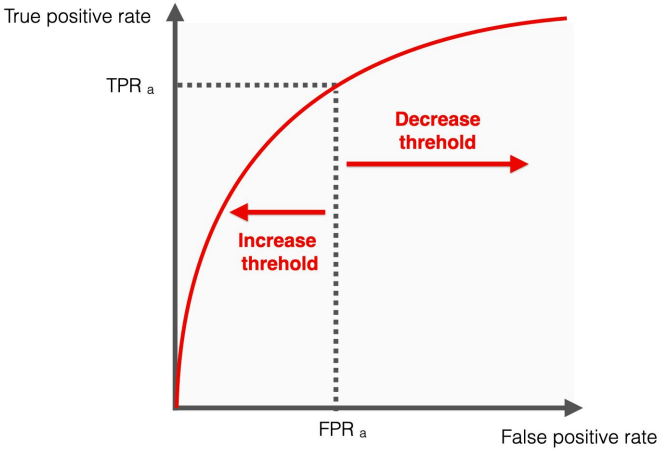
Threshold classifiers and ROC curve

ROC (Receiver Operating Characteristic) curve illustrates the tradeoff between the TPR and FPR.
For every threshold, we calculate TPR and FPR and plot them on one chart.

TPR/ Recall $= \dfrac{\text{True Positives}}{\text{True Positives + False Negatives}}$

FPR $= \dfrac{\text{False Positives}}{\text{False Positives + True Negatives}}$





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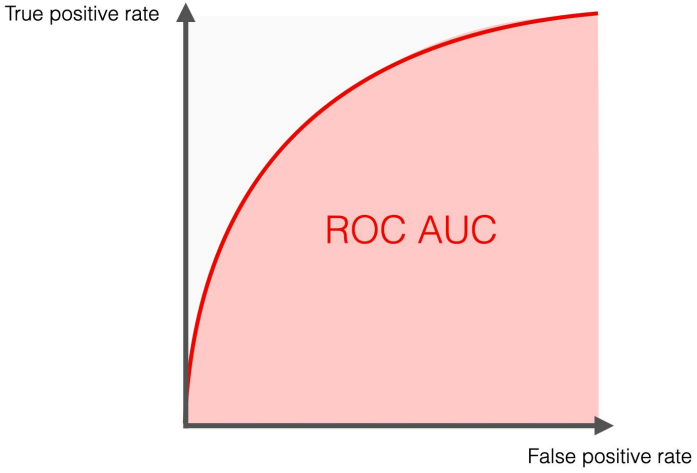
Threshold classifiers and ROC AUC metric

ROC Area Under the Curve

TPR/ $$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

$$\text{FPR} = \frac{\text{False Positives}}{\text{False Positives} + \text{True Negatives}}$$



A higher ROC AUC indicates better performance. A perfect model would have an AUC of 1.

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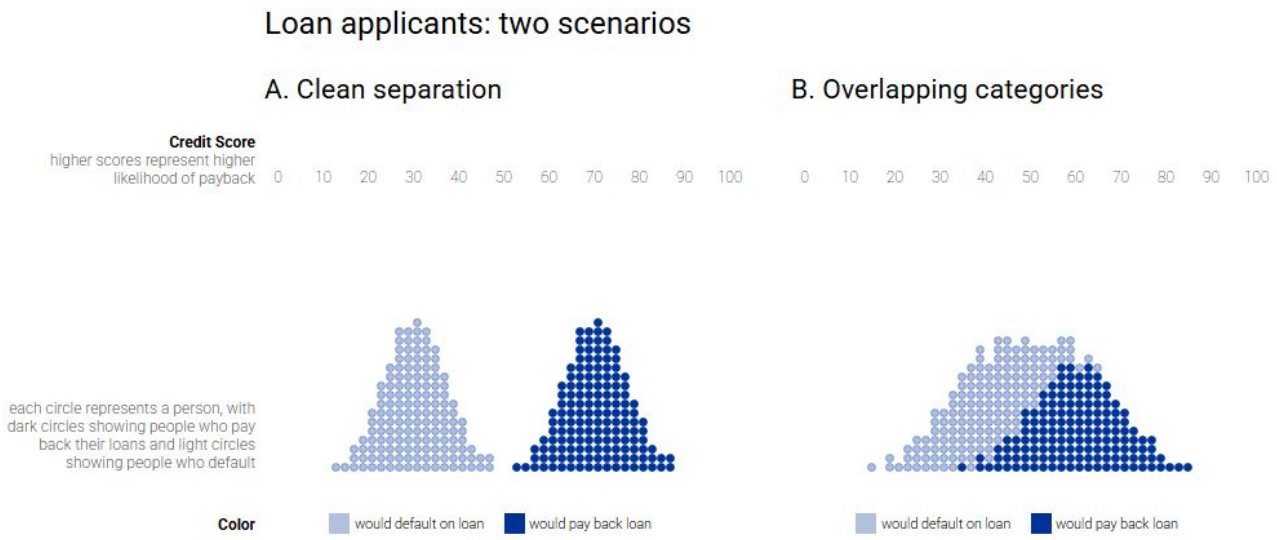
Address AI Bias with Fairness Criteria

Attacking discrimination with smarter machine learning

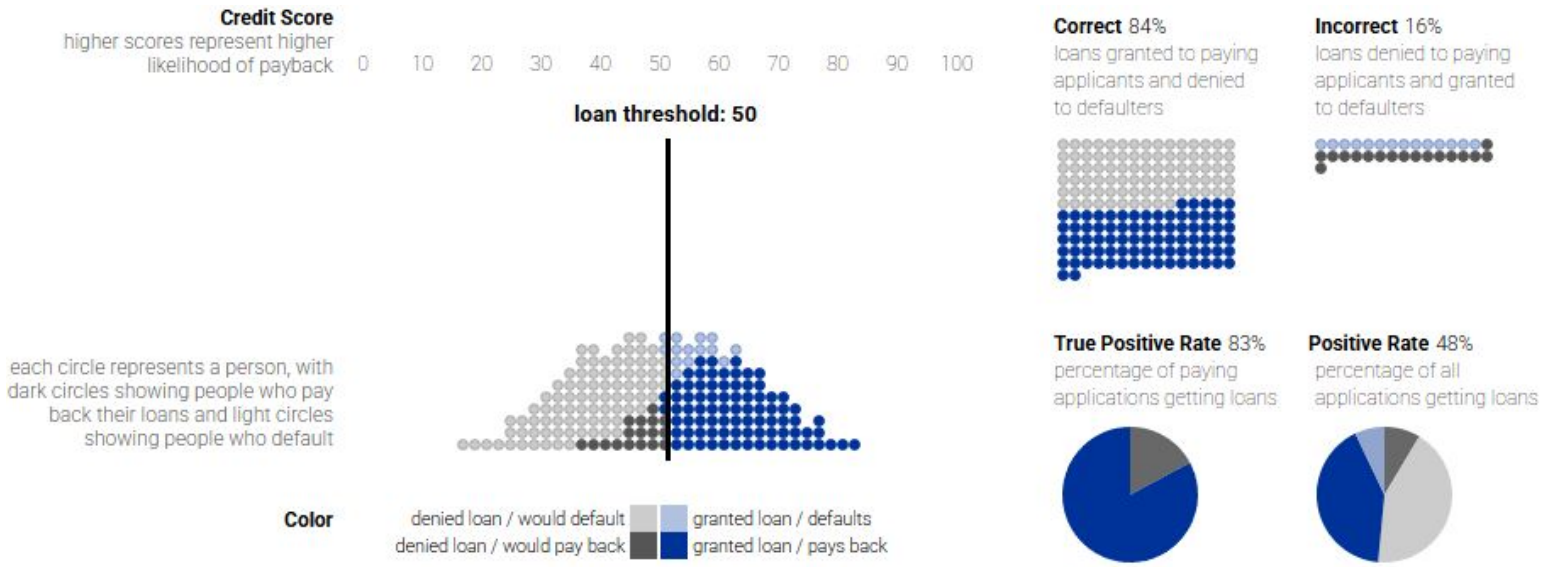
Threshold classifiers



https://research.google.com/bigpicture/attacking-discrimination-in-ml/

Threshold classifiers

Suppose, the bank picks a particular cut-off, or threshold, and people whose credit scores are below it are denied the loan, and people above it are granted the loan.

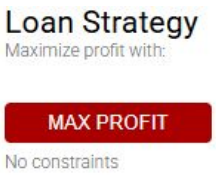


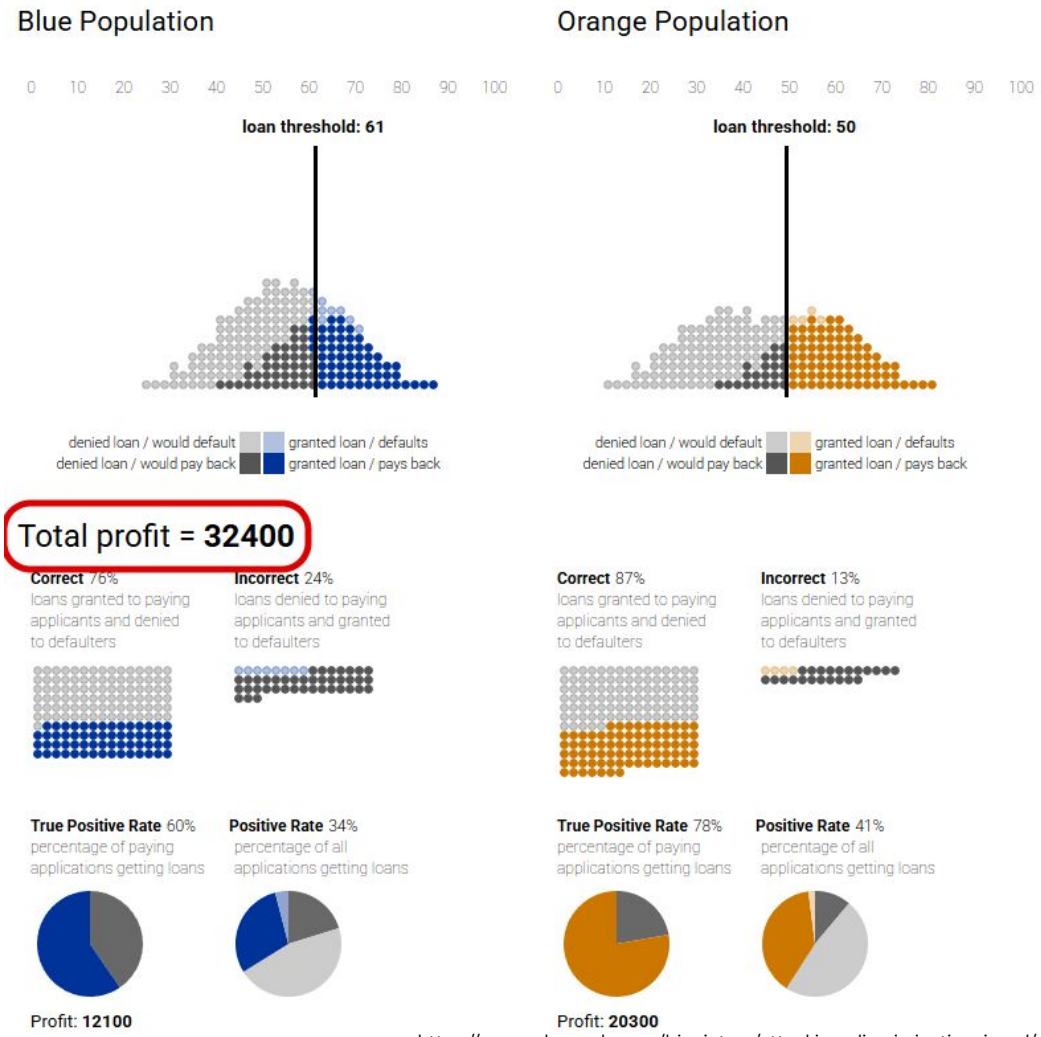
Picking a threshold requires some tradeoffs. Too low, and the bank gives loans to many people who default. Too high, and many people who deserve a loan won't get one.

https://research.google.com/bigpicture/attacking-discrimination-in-ml/

two groups of people, "blue" and "orange."
loans are subject to the following rules:

- A successful loan makes $300
- An unsuccessful loan costs $700
- Everyone has a credit score between 0 and 100





The most profitable, since there are no constraints.
But **the two groups have different thresholds, meaning they are held to different standards.**

https://research.google.com/bigpicture/attacking-discrimination-in-ml/

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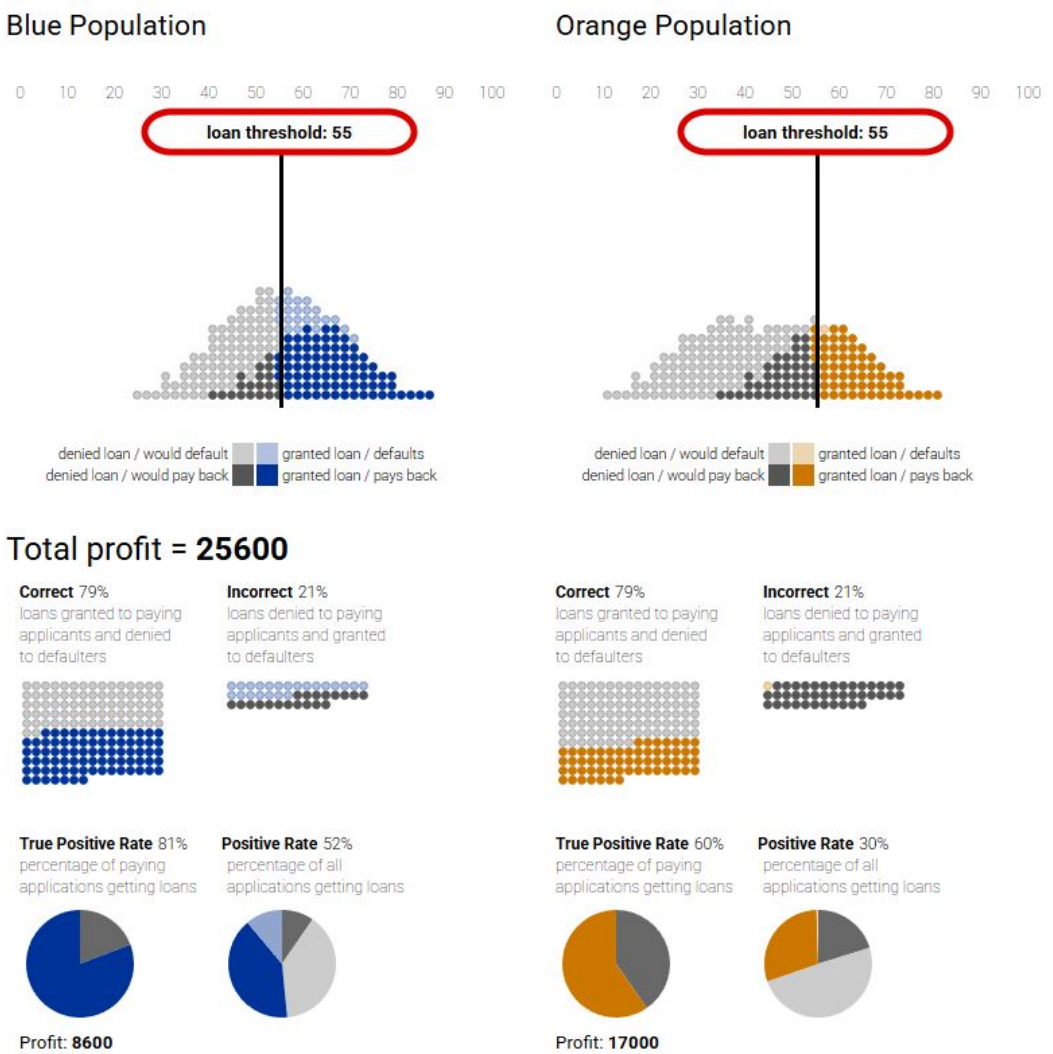
GROUP UNAWARE

Blue and orange thresholds are the same



Both groups have the same threshold, but **the orange group has been given fewer loans overall.**
Among people who would pay back a loan, the orange group is also at a disadvantage.

https://research.google.com/bigpicture/attacking-discrimination-in-ml/

is group unawareness really fair in real life?

Ending Affirmative Action Harms Diversity Without Improving Academic Merit

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EAAMO '24, https://dl.acm.org/doi/pdf/10.1145/3689904.3694706

is group unawareness really fair in real life?

Some proxy measures or attributes are correlated with sensitive characters.

Zip code, college, or an applicant’s name may have a correlation with caste and religion.

two groups of people, "blue" and "orange."
loans are subject to the following rules:

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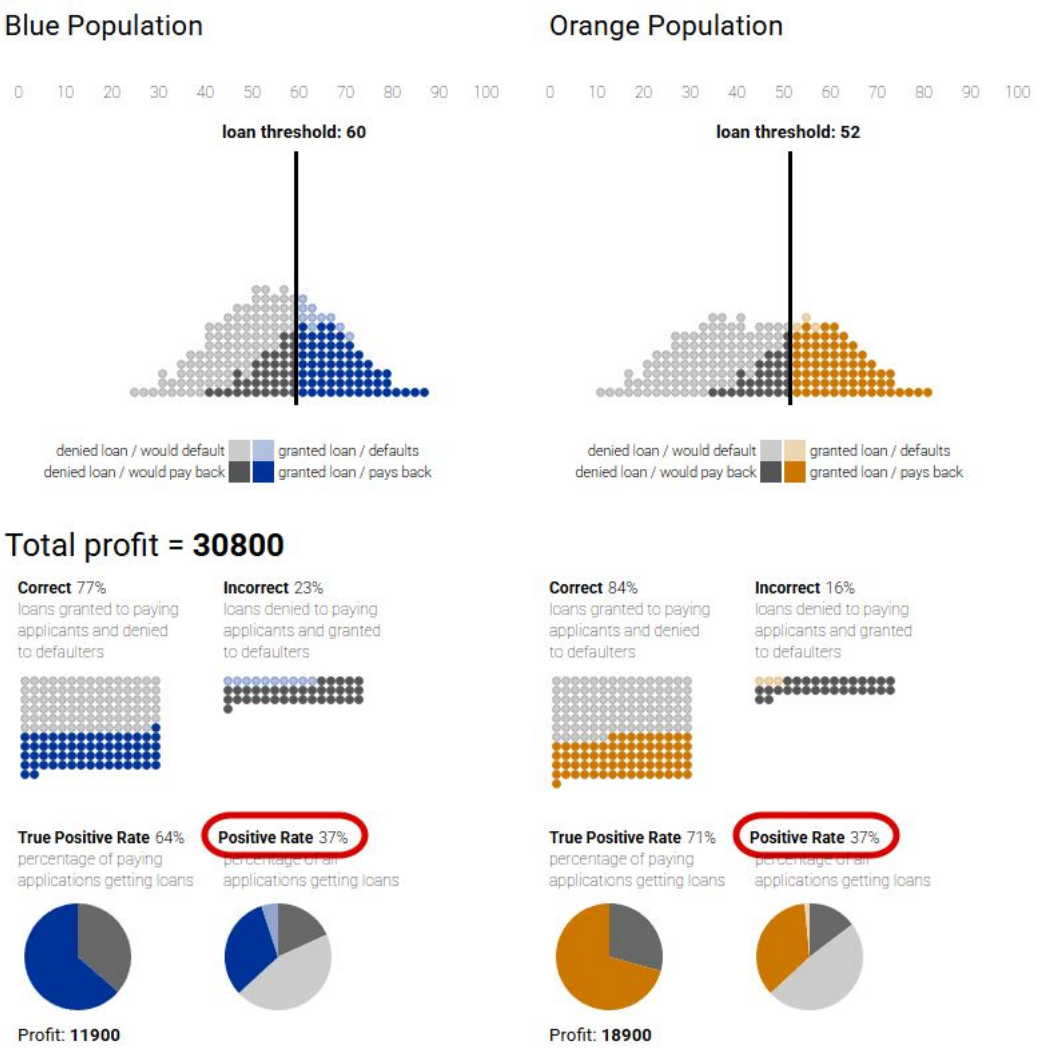
DEMOGRAPHIC PARITY

Same fractions blue / orange loans



The number of loans given to each group is the same,
but **among people who would pay back a loan, the blue group is at a disadvantage**.

https://research.google.com/bigpicture/attacking-discrimination-in-ml/

two groups of people, "blue" and "orange."
loans are subject to the following rules:

- A su
- An
- Eve
  and

The nu
group is the same,
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two groups of people, "blue" and "orange."
loans are subject to the following rules:

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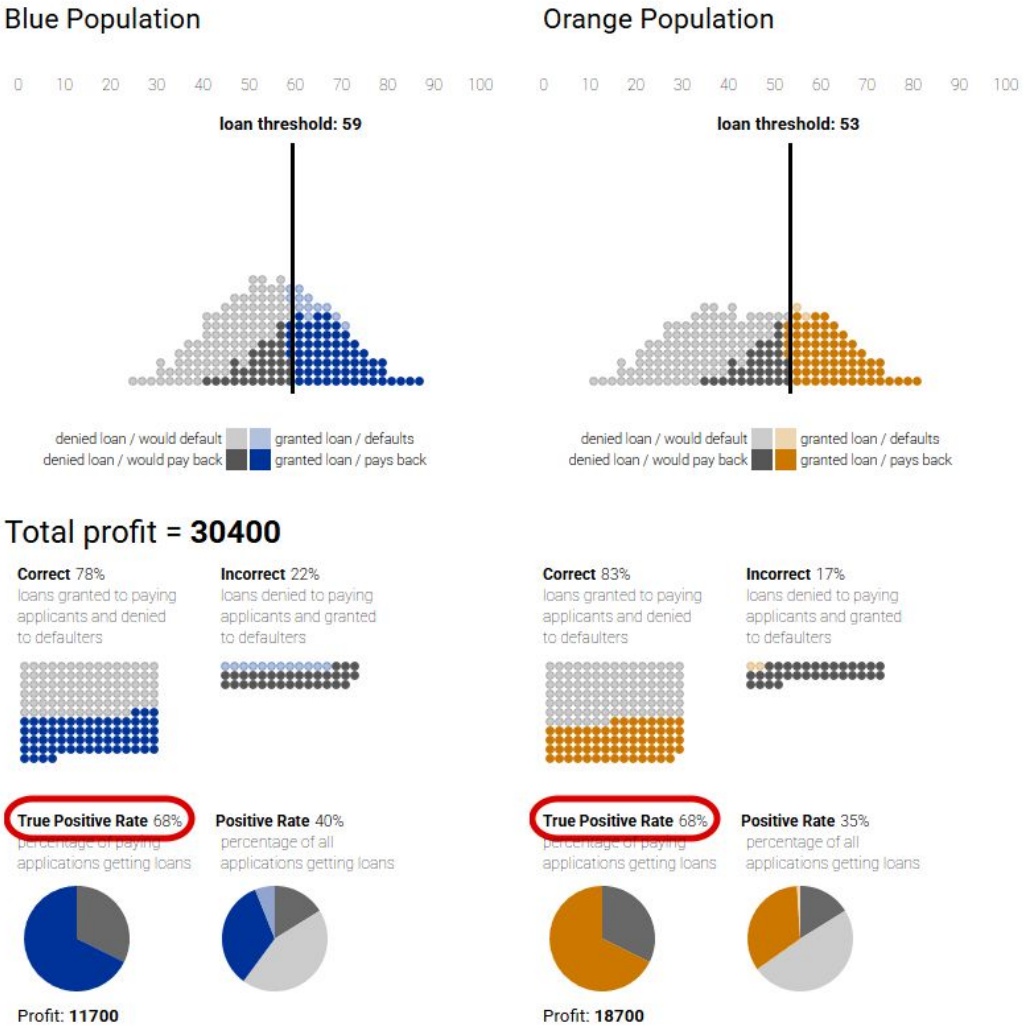
EQUAL OPPORTUNITY

Same fractions blue / orange loans to people who can pay them off



Among people who would pay back a loan, blue and orange groups do equally well. This choice is almost as profitable as demographic parity, and about as many people get loans overall.

https://research.google.com/bigpicture/attacking-discrimination-in-ml/

Threshold classifiers: Equal accuracy

same percentage of correct judgment in both groups.

Equal opportunity

Group A: 66% true positive rate: 4/(4+2)
Group B: 66% true positive rate: 2/(1+2)

GROUP A

TRUE \ PREDICTED	Deny	Approve
Deny	3	1
Approve	2	4

GROUP B

TRUE \ PREDICTED	Deny	Approve
Deny	6	1
Approve	1	2

Next, we can see how the confusion matrices might look for **equal accuracy** fairness. For each group, the model was 80% accurate.

Equal accuracy

Group A: 80% accurate: (6+2)/10
Group B: 80% accurate: (4+4)/10

GROUP A

TRUE \ PREDICTED	Deny	Approve
Deny	2	1
Approve	1	6

GROUP B

TRUE \ PREDICTED	Deny	Approve
Deny	4	2
Approve	0	4

https://www.kaggle.com/code/alexisbcook/ai-fairness

Question: In a medical screening (for a rare disease) model deployed across demographic groups with different base rate (actual proportion of the disease), would you target

1. Equal opportunity (equal TP), or
2. Equal FN, or
3. Demographic Parity (equal TP+FP) across groups?

Explain which harm you are minimizing and why. What if you target the other two factors?